On the Path Dependence of Derivatives via Monte Carlo Methods

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Abstract

Option prices that obey the Black–Scholes differential equations can be evaluated analytically in only a handful of cases. Monte Carlo simulations are efficient numerical methods that scale well with the number of dimensions and are easy to implement on a computer. In this project, we focus on the Monte Carlo pricing of options for which no analytical solution exists. This is the case for American options and more exotic options that depend on the path taken by the underlying asset. In this project, I will consider American options and an Americanized arithmetic-average Asian option.

1. Methods

American options incorporate early exercise and path dependence but we can still characterize the price by S_t and t. However early excercise makes the American options an optimization problem. Despite its dependence on the path American options with a European-like payoff are weakly path dependent, that is we do not require an additional variable to describe its price [Wilmott(2006)]. Let $V(S,t)$ denote the price of an option with a vanilla-like payoff. For concreteness, let us consider a European put with payoff

$$G(S,t) = \max(E - S)\,,$$

where E is the strike price and S is the stock price. The price of the option at present time ($t = 0$) is defined as

$$V_0 = \sup_{t \in \mathcal{T}_{[0,T]}} \mathbb{E}^{\mathbb{Q}}\Big[e^{-rt} G(S_t, t)\Big]\,,$$

where $\mathbb{Q}$ is a risk-neutral measure, r is the risk-free rate, and $\mathcal{T}_{[0,T]}$ is the set of stopping times with values in $[0,T]$. The optimal stopping time solves this supremum. In practice, we discretize $[0,T]$ to a grid $0 = t_0, t_1, \ldots, t_N = N$ and search over stopping times taking values in this grid.

1.1 Longstaff–Schwartz algorithm

The Monte-Carlo simulation of this option follows the Longstaff–Schwartz least squares method (LSM) [Longstaff and Schwartz(2001)]. The method can be summarized as follows.

1. Simulate M paths forward from S_0 to T on a grid of $N+1$ dates; record the matrix $S_{m,n}$ of all simulated prices.

2. Starting at T and moving backward in time, at each date t_n with $0 < n < N$: identify the paths that are in-the-money (ITM) at t_n; regress the (already-computed, discounted) continuation values on a set of basis functions of S_{t_n}; this regression gives an approximation $\hat{C}_n(S) = \sum_k \beta_k \psi_k(S)$ of the continuation value $C_n(S) = \mathbb{E}^{\mathbb{Q}}[V_{n+1} \mid S_{t_n} = S]$.

3. At each ITM path, compare the immediate exercise payoff $G(t_n, S_{m,n})$ with the regression-estimated continuation value $\hat{C}_n(S_{m,n})$. If the exercise payoff exceeds the continuation value, mark the path as "exercise at t_n" and update its value to the exercise payoff; otherwise the path carries forward its existing (discounted) future value.

4. After the backward sweep, the time-0 price is the average of the path-wise discounted values. The choice of basis functions is important but not critical. We will use a power-like regression approach and use $1, S, S^2, \ldots$ as our basis.

2. Results

2.1 American put

We have worked out the calculation of reference obtaining the results in table 1. The LSM result are compared against the results obtained by the Finite Differences Method (FDM). See also fig. 1.

S	σ	T	European	American (FDM)	American (LSM)	(s.e.)	Early Ex. (FDM)	Early Ex. (LSM)	Diff Ex. value
36	0.200	1.000	3.844	4.479	4.477	0.008	0.635	0.633	0.002
36	0.200	2.000	3.763	4.837	4.837	0.010	1.074	1.074	0.001
36	0.400	1.000	6.711	7.107	7.084	0.012	0.396	0.372	0.023
36	0.400	2.000	7.700	8.512	8.474	0.015	0.812	0.774	0.039
38	0.200	1.000	2.852	3.247	3.240	0.007	0.395	0.388	0.007
38	0.200	2.000	2.991	3.739	3.732	0.009	0.748	0.741	0.007
38	0.400	1.000	5.834	6.153	6.143	0.011	0.318	0.309	0.009
38	0.400	2.000	6.979	7.673	7.652	0.014	0.694	0.673	0.021
40	0.200	1.000	2.066	2.309	2.312	0.007	0.242	0.245	-0.003
40	0.200	2.000	2.356	2.876	2.888	0.009	0.521	0.532	-0.012
40	0.400	1.000	5.060	5.316	5.301	0.012	0.256	0.241	0.015
40	0.400	2.000	6.326	6.921	6.920	0.014	0.595	0.594	0.001
42	0.200	1.000	1.465	1.611	1.623	0.008	0.146	0.158	-0.012
42	0.200	2.000	1.841	2.204	2.217	0.010	0.362	0.376	-0.013
42	0.400	1.000	4.379	4.586	4.579	0.014	0.207	0.200	0.007
42	0.400	2.000	5.736	6.248	6.252	0.015	0.512	0.516	-0.004
44	0.200	1.000	1.017	1.104	1.119	0.008	0.087	0.102	-0.015
44	0.200	2.000	1.429	1.681	1.691	0.010	0.252	0.262	-0.010
44	0.400	1.000	3.783	3.951	3.956	0.016	0.168	0.173	-0.005
44	0.400	2.000	5.202	5.644	5.644	0.017	0.442	0.442	0.000

Table 1: Pricing using LSM. Simulations were performed using $100\,000$ paths, assuming the options are exercisable 50 times a yer. Withing the quoted error this table matches LSM paper.

2.2 Arithmetic Asian option with early exercise

The payoff of this options reads

$$V(\{S_t\}) = \max(E - A_{t_n}, 0), \quad A_{t_n} = \frac{1}{n+1} \sum_{k=0}^{n} S_{t_k} \,,$$

exercisable at any $t_n \in \{ t_1, t_2, \ldots, t_N \}$. Unlike the American put, here we need an additional variable to characterize the option. In this sense, the option is strongly path-dependent. We also compare the MC prices against the FDM prices. See fig. 1.

S	σ	T	European	American Asian (FDM)	(err num)	American Asian (LSM)	(s.e.)
36	0.200	1.000	3.844	4.604	0.000	4.655	0.006
36	0.200	2.000	3.763	4.748	0.001	4.779	0.007
36	0.400	1.000	6.711	5.992	0.001	5.917	0.010
36	0.400	2.000	7.700	6.718	0.003	6.645	0.012
38	0.200	1.000	2.852	2.838	0.000	2.803	0.006
38	0.200	2.000	2.991	3.096	0.001	3.055	0.008
38	0.400	1.000	5.834	4.606	0.001	4.509	0.010
38	0.400	2.000	6.979	5.482	0.003	5.401	0.012
40	0.200	1.000	2.066	1.546	0.000	1.499	0.006
40	0.200	2.000	2.356	1.903	0.002	1.861	0.008
40	0.400	1.000	5.060	3.477	0.001	3.377	0.011
40	0.400	2.000	6.326	4.449	0.003	4.379	0.013
42	0.200	1.000	1.465	0.776	0.000	0.743	0.006
42	0.200	2.000	1.841	1.136	0.001	1.110	0.008
42	0.400	1.000	4.379	2.587	0.001	2.500	0.012
42	0.400	2.000	5.736	3.597	0.003	3.537	0.014
44	0.200	1.000	1.017	0.363	0.000	0.343	0.004
44	0.200	2.000	1.429	0.662	0.001	0.644	0.006
44	0.400	1.000	3.783	1.901	0.001	1.828	0.011
44	0.400	2.000	5.202	2.898	0.003	2.845	0.014

Table 2: Pricing table of an arithmetic Asian option

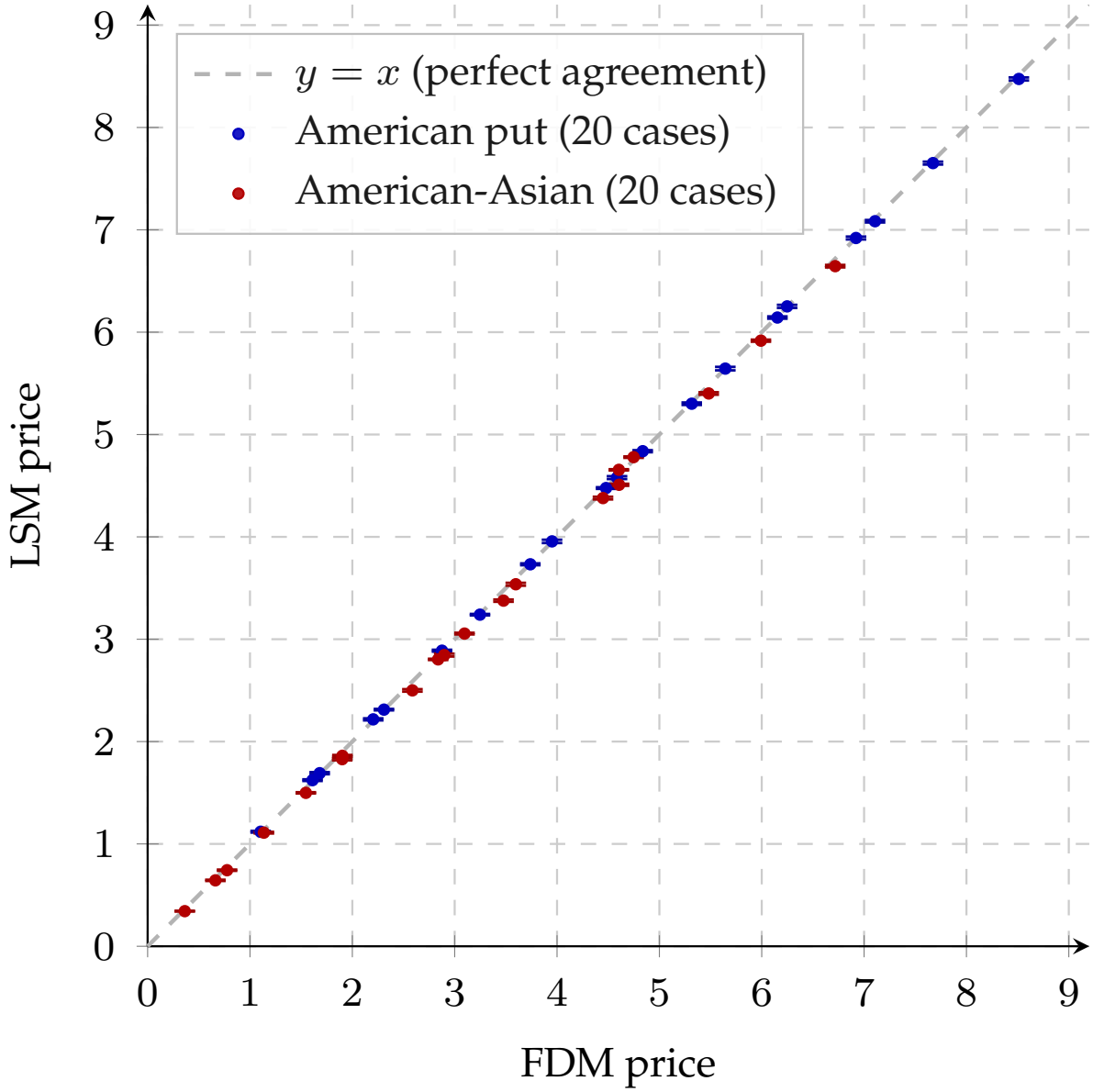


Figure 1: LSM Monte Carlo price vs. finite-difference (FDM) price, for all 20 American-put cases (blue circles, Table 1) and all 20 American-Asian cases (red squares, Table 2). Vertical error bars show the LSM standard error ($M = 100{,}000$ paths, $N = 50$ exercise dates). The dashed grey line is $y = x$ (perfect agreement). All 40 points lie within ~ 0.1 of the diagonal and well within their standard-error bars, confirming that the two independent methods agree across the full parameter grid.

2.3 Comparison between American and Asian with early exercise

Early exercise generally speaking add value to the option. Therefore American options are usually more expensive than Vanilla options. On the other hand Asian option takes the average of the underlying over some prescribed period and this decreases its price when compared against a similar Vanilla. Thus, we have the inequalities

$$V^{\text{Vanilla}} \leq V^{\text{American}}, \qquad V^{\text{Asian}} \leq V^{\text{Vanilla}}$$

The americanization of the Asian option would suggest that a similar inequality holds for the americanized asian option. However, this is not always the case as we can see in Table 3. The expected behavior is violated in a few cases of low volatility and longer exercise time. This is consistent with the conclusion of G.L. Ye in ref. [Ye(2008)].

3. Conclusion

In this project, we have studied path-dependent options using Monte Carlo simulations and the Longstaff–Schwartz (LSM) approach. We have reproduced the numerical results presented in the original LSM paper and extended the analysis to the pricing of an Americanized arithmetic-average Asian option. Interestingly, the price of the latter is not necessarily smaller than that of the corresponding European option.

S	sigma	T	American Asian (LSM)	European	American (LSM)	American-(American Asian)	European-(American Asian)
36	0.200	1.000	4.655	3.844	4.477	-0.178	-0.810
36	0.200	2.000	4.779	3.763	4.837	0.058	-1.016
36	0.400	1.000	5.917	6.711	7.084	1.166	0.794
36	0.400	2.000	6.645	7.700	8.474	1.829	1.055
38	0.200	1.000	2.803	2.852	3.240	0.437	0.049
38	0.200	2.000	3.055	2.991	3.732	0.676	-0.065
38	0.400	1.000	4.509	5.834	6.143	1.635	1.325
38	0.400	2.000	5.401	6.979	7.652	2.252	1.578
40	0.200	1.000	1.499	2.066	2.312	0.813	0.568
40	0.200	2.000	1.861	2.356	2.888	1.027	0.494
40	0.400	1.000	3.377	5.060	5.301	1.924	1.683
40	0.400	2.000	4.379	6.326	6.920	2.541	1.947
42	0.200	1.000	0.743	1.465	1.623	0.879	0.721
42	0.200	2.000	1.110	1.841	2.217	1.107	0.731
42	0.400	1.000	2.500	4.379	4.579	2.079	1.879
42	0.400	2.000	3.537	5.736	6.252	2.715	2.198
44	0.200	1.000	0.343	1.017	1.119	0.775	0.674
44	0.200	2.000	0.644	1.429	1.691	1.047	0.786
44	0.400	1.000	1.828	3.783	3.956	2.129	1.955
44	0.400	2.000	2.845	5.202	5.644	2.799	2.357

Table 3: Comparison of European, American, and Asian with early exercise

References

[Longstaff and Schwartz(2001)] F. A. Longstaff and E. S. Schwartz. Valuing American options by simulation: A simple least-squares approach. *Review of Financial Studies*, 14(1):113–147, 2001. doi: 10.1093/rfs/14.1.113.

[Wilmott(2006)] P. Wilmott. *Paul Wilmott on Quantitative Finance*. John Wiley & Sons, Chichester, 2nd edition, 2006. 3 volumes; first edition 1998.

[Ye(2008)] G. L. Ye. Asian options versus vanilla options: a boundary analysis. *Journal of Risk Finance*, 9(2):188–195, 2008. doi: 10.1108/15265940810853931. Shows that the commonly held belief that an Asian option is always cheaper than its European vanilla counterpart is false; identifies the parameter boundary where the ordering reverses.